

Reinforcement Learning for Simple Adaptive Robot Behavior

Topic 6 / Track 8 — Learning, Personalization & Adaptation in HRI

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Interactive & Social AI for Humanoid Robots

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Framing & Foundations

What this topic is — and where it sits

Our assigned topic: Reinforcement Learning for Simple Adaptive Robot Behavior.

- Realized through **Topic 6: Learning in HRI** (RL vs. SL, learning from demonstration, personalization).
- Anchored in the seminar's **Track 8: Personalization & Adaptation** paper set.
- Implemented under **Project Track D: Adaptive & Learning Systems** on the **NAO** robot.

The one-sentence thesis

A robot can *learn online*, from a person's social signals as a *reward*, to adapt simple social behaviors — formalized as a Markov Decision Process and solved with lightweight RL.

Robots, social robots, and HRI

Robot: an embodied machine that *senses*, *computes*, and *acts* on the world.

Social robot: interacts with people following social rules, using human-like cues.

HRI: the interdisciplinary field — robotics, AI, psychology, design — studying how people and robots interact.

Core challenges

- Embodiment — the body shapes interaction
- Autonomy — how much the robot decides
- Trust — whether people rely on it

Modalities & signals

- Speech, gesture, gaze, **proxemics** (space)
- **Social signals** (gaze, repositioning, touch) → our *reward*

Learning Paradigms

Three ways a machine can learn

Supervised

- Learn input → correct label from examples.
- Needs large *pre-labeled* data; trained *offline*.
- E.g. emotion classifier from labeled images.

Unsupervised

- Find structure in *unlabeled* data.
- Clustering, dimensionality reduction.
- E.g. group users by interaction style.

Reinforcement

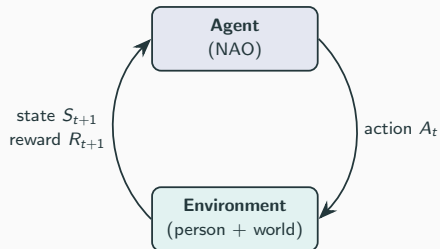
- Learn by *trial and error* from a *reward*.
- No labels; adapts *online*.
- E.g. adjust distance if the person leans in / backs away.

Why not just supervised learning?

The “correct” social behavior is person- and context-specific, unknown in advance, and shifts *during* interaction. Supervised learning needs a labeled answer up front and cannot adapt mid-interaction. RL learns the answer *from* the interaction.

Reinforcement Learning in Depth

The RL loop: agent, environment, reward



At each step t the agent:

1. observes state S_t ,
2. takes action A_t via policy $\pi(a \mid s)$,
3. receives reward R_{t+1} ,
4. lands in state S_{t+1} .

Goal: find π maximizing expected *discounted return*

$$G_t = \sum_{k \geq 0} \gamma^k R_{t+k+1}, \quad \gamma \in [0, 1).$$

Value $Q(s, a)$ = expected return from taking a in s .

Formalized as an MDP $\langle S, A, P, R, \gamma \rangle$: the next state depends only on current (s, a) — the *Markov property*.
[Sutton2018]

Q-learning: the workhorse for simple adaptive behavior

Idea: learn the value of each (state, action) pair directly — no model of the world needed.

Q-learning update (Watkins & Dayan, 1992)

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[\underbrace{r + \gamma \max_{a'} Q(s', a')}_{\text{target}} - Q(s, a) \right]$$

α : learning rate γ : discount r : observed reward

Exploration vs. exploitation

- Must try new actions *and* use known-good ones.
- ϵ -greedy: random action w.p. ϵ , else $\arg \max_a Q(s, a)$; decay ϵ over time.

Tabular vs. approximation

- *Tabular*: one value per (s, a) — perfect for small, discrete spaces (our case).
- Large/continuous $\rightarrow$ function approximation (Deep Q-Networks).

Policy gradients & learning from human reward

Policy gradient (REINFORCE)

- Directly tune a parameterized policy π_θ toward higher reward.
- Natural fit for *continuous* action parameters: distance, speed, timing.
- Used by Mitsunaga et al. to adapt proxemics, gaze & timing on a social robot. [Mitsunaga2005]

Interactive RL / human feedback

- The *human* supplies the reward signal.
- TAMER: non-experts shape an agent via approval / disapproval. [Knox2009]
- Cuts the number of trials needed — crucial on hardware.

The hardware reality (a recurring theme)

Every real trial costs *time*, *battery*, and *joint wear*. So **sample efficiency** matters, and **sim-to-real transfer** (the “reality gap”) is a genuine risk: policies that work in simulation can fail on the physical robot.

Learning from Demonstration (LfD) — a complement to RL

Teach by showing, not by coding or rewarding.

[Argall2009]

- Intuitive for non-experts — people teach by demonstrating.
- **Kinesthetic teaching**: set joint stiffness to zero, physically move NAO's limb, record the joint trajectory, replay it.

Why pair LfD with RL?

A demonstration gives a good *starting policy*; RL then *refines* it through interaction — fixing RL's slow cold-start and poor sample efficiency.

Personalization & Adaptation

Personalization in long-term HRI

Why adapt at all?

- Short-term studies hide the hard part: sustaining engagement over many sessions. [Irfan2019]
- Adaptation counters the **novelty effect** — interest fades once a robot stops being new.
- Built on *user modeling + incremental / lifelong learning*. [Irfan2021]

Static vs. dynamic

- *Static*: fix a profile once (a questionnaire).
- **Dynamic**: adjust in real time to live social signals — our RL approach.

The ethical edge: entertainment vs. manipulation [Pollmann2023]

The same personalization that improves user experience can be used to *steer* a person against their own autonomy. Adaptive systems need explicit ethical guardrails — a point we will state up front, not bolt on.

Literature Review

Anchor papers (Track 8 set)

- **Irfan et al. (2019), Personalization in Long-Term HRI.** Establishes the premise: adaptation/personalization are *necessary* to sustain engagement and build trust. [\[Irfan2019\]](#)
- **Irfan et al. (2021), Lifelong Learning & Personalization (LEAP-HRI).** Robots “in the wild” must learn new concepts incrementally without forgetting. [\[Irfan2021\]](#)
- **Tozadore et al. (2018), Towards Adaptation & Personalization in Task-Based HRI.** A concrete personalization framework on a humanoid; classifies user signals to adapt activities and prolong interaction. [\[Tozadore2018\]](#)
- **Pollmann et al. (2023), Entertainment vs. Manipulation.** User-centered + ethical analysis of a personalized robot; personalization helps UX but risks manipulation. [\[Pollmann2023\]](#)

Added papers — the RL “how” the anchors leave open

- **Mitsunaga et al. (2005/2008), Robot Behavior Adaptation via Policy-Gradient RL.** The seminal method: a social robot uses policy-gradient RL to adjust *interaction distance*, *gaze meeting*, and *motion speed/timing*, reading subconscious human comfort signals as reward; adapts to individual preferences across 12 subjects. [\[Mitsunaga2005\]](#)
- **Patompak et al. (2020), Learning Proxemics for Personalized HRI.** Builds a Social Force Model via fuzzy inference and tunes it with RL so the robot learns each person's comfortable interaction distance; validated in simulation and on a real robot. [\[Patompak2020\]](#)

Rationale: the anchors explain *why* and *what* to personalize; these two supply concrete *RL mechanisms* for adapting social behavior — with the proxemics formulation that maps cleanly onto NAO's sonar + tactile sensors.

Methodological critique & research gaps

- **Reality gap:** fast convergence in simulation, but sensor noise and dynamics differ on hardware.
- **Sample efficiency:** real trials are slow and cause wear — few-shot learning, human reward, and LfD warm-starts help.
- **Reward specification:** “social reward” is noisy and hard to define; bad rewards yield awkward behavior.
- **Static vs. dynamic preference:** treating preference as fixed fails long-term.
- **Evaluation:** over-reliance on subjective post-hoc surveys; need to *combine* objective sensor metrics with subjective scales.
- **Generalization & reproducibility:** per-user policies may not transfer; HRI lacks standardized metrics/protocols.

These gaps are our **Q&A defense** — for each we will hold a one-line position grounded in the literature.

The NAO Platform

NAO V6 — our research platform

Hardware [Aldebaran]

- 58 cm, ~5.6 kg, **25 degrees of freedom**
- Dual 5 MP cameras; 4 microphones (sound localization)
- **Sonar** distance sensors: ~25–255 cm
- **Tactile** sensors (head + hands); foot force sensors; IMU; joint encoders
- Battery: **45–90 min**; limited onboard compute

Software (NAOqi SDK)

- ALMotion — joints, posture, stiffness, walking
- ALMemory / ALSensors — sensor blackboard
- ALSonar — distance; ALTextToSpeech — speech
- **Choregraphe** IDE (with a *virtual* NAO)
- Python / C++ SDKs; ROS support

Open question for the instructors

Which NAO version is on campus, and may we run trials on the physical robot? Until confirmed, we prototype the full RL loop in **simulation** / **mocked NAOqi**.

Project Pitch

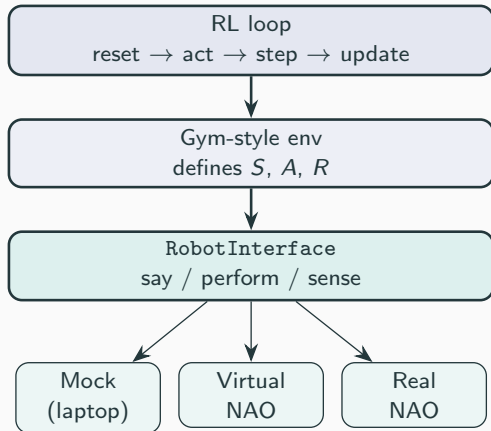
Three candidate experiments (all NAO + RL, 2-person, 2–4 weeks)

Option	Adaptive behavior	Reward signal	Baseline
(A) Greeting / social distance	Approach distance + greeting intensity	Tactile “approve” + sonar distance change	Fixed scripted greeting
(B) Tutoring / pace	Speech rate, pauses, difficulty	Answer correctness + engagement	Fixed-pace tutor
(C) Engagement max.	Pick from behavior repertoire (joke, gesture, question)	Change in gaze + proximity	Random / fixed rotation

All three are deliberately kept **open**. They share one RL loop — only the state, actions, and reward wiring differ. **We are not pre-committing**: we would like the discussion to pick a direction (from these, or a new one you suggest).

Shared architecture — why any direction is one line away

Every candidate experiment uses the *same* RL loop; only the state/action/reward bindings change. An abstract robot interface lets the identical loop run on laptop, simulator, or real NAO.



- **Already built & running** on the laptop mock — this is our “implementation has begun”.
- **Sensor noise injected** in the mock to harden the policy and shrink the eventual reality gap.
- Swapping to a NAO is literally:

```
# robot = MockRobot()  
robot = NaoRobot(ip=...)  
— env and agent code unchanged.
```

Common NAOqi plumbing (excerpt of working code)

The seam that makes the swap trivial. The mock contains a simulated person so the loop runs with *no hardware and no NAOqi SDK* today.

```
class RobotInterface(ABC):          # shared by mock / sim / real
    def perform(self, action): ...   # 'closer', 'back', 'intensity_up'...
    def sense(self) -> Signals: ...  # distance, touch, present

class MockRobot(RobotInterface):     # laptop + hidden person model
    def sense(self):
        return Signals(distance=self._d + noise,
                        touch=self._person_approves(),
                        present=self._present)

class NaoRobot(RobotInterface):      # real NAO via NAOqi
    def sense(self):
        d = self.memory.getData("../US/Left/Sensor/Value")
        touch = self.memory.getData("FrontTactilTouched") > 0.5
        # ... ALSonar + tactile + ALFaceDetection
```

Result so far: on the illustrative greeting task, the learned policy beats a scripted baseline by $\sim +9$ reward, discovering “don’t crowd; settle in the personal zone” from touch signals alone — hosted on GitHub.

Worked example (illustrative) — a greeting MDP

State S (discretized)

- Distance bin: intimate / personal / social (from sonar)
- Current greeting intensity level

Action A

- move closer / move back
- increase / decrease intensity / hold

Reward R

- + head touch (approve), sustained gaze, user stays
- – user steps back (sonar), no touch, leaves

Algorithm

- Tabular Q-learning, ϵ -greedy (decaying ϵ)
- Small discrete space $\Rightarrow$ converges in few episodes

A small, discrete MDP we can defend in detail — and the template we would re-instantiate for whichever direction the committee chooses.

Study design

- **Within-subject**: each participant meets the scripted *and* the learned robot (counterbalanced order).
- Higher power; controls individual differences.

Objective metrics

- Convergence speed (trials to stable policy)
- Final interaction distance, approach latency
- Engagement duration

Subjective metrics

- **Godspeed** questionnaire (likeability, perceived safety, perceived intelligence)
- Comfort / trust survey

The point

- *Triangulate*: pair sensor metrics with subjective scales — directly answering the field's main critique.

Implementation plan — simulation first

1. **Mock / Gym environment.** Stub the NAOqi proxies (fake ALMotion, ALMemory/sonar, tactile); encode the $\langle S, A, R \rangle$ MDP; verify Q-learning converges.
2. **Virtual NAO.** Swap mocks for Choregraphe's virtual robot or Webots (ships a NAO model; NAOqi/ROS support).
3. **Real NAO (if available).** Fine-tune for a small number of trials; monitor joint heat / battery.

Engineering hygiene

Code hosted on **GitHub** from day one; modular hardware-abstraction layer so the same RL loop runs against mock, simulator, or hardware unchanged.

Plan & References

Timeline to presentation & delivery

- **Now** → **submission**: lock this draft after instructor feedback; divide papers for deep reading.
- **This week**: stand up the mock Gym environment + tabular Q-learning; confirm convergence; push to GitHub.
- **Next**: move to virtual NAO (Choregraphe/Webots); integrate sonar + tactile reward.
- **Pending instructor reply**: confirm NAO version & hardware access → decide on real-robot trials.
- **Final**: within-subject mini study, combine objective + Godspeed metrics, write thesis-style report.

Our specific asks are on the next slide.

Questions for the committee — resources & scope

Hardware & access

- Which **NAO version** is on campus (V5 / V6), how many units, and how do we *book* time?
- May we run **participant trials** on the physical robot, and in which lab / hours?

Compute

- Our tabular Q-learning runs on a **laptop CPU** — no GPU needed.
- *Only if* we extend to **deep RL** or onboard vision would we need GPU / compute — is any available?

Study logistics

- May we **recruit fellow students** as participants? Is there a pool or sign-up?
- What **ethics / consent** process is required for a user study on campus?

Scope decision

- Which **direction** should we pursue — one of A / B / C, or a new one you suggest?
- Any constraints we should design around now rather than later?

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- [3] D. C. Tozadore et al., “Towards Adaptation and Personalization in Task Based on Human-Robot Interaction,” *LARS/SBR/WRE*, 2018, pp. 383–389. (IEEE Xplore doc. 8588581; DOI to verify.)
- [4] K. Pollmann, W. Loh, N. Fronemann, D. Ziegler, “Entertainment vs. manipulation: Personalized human-robot interaction,” *Technological Forecasting and Social Change*, vol. 189, art. 122376, 2023. DOI: 10.1016/j.techfore.2023.122376.

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- [6] P. Patompak, S. Jeong, I. Nilkhamhang, N. Y. Chong, “Learning Proxemics for Personalized Human–Robot Social Interaction,” *Int. J. of Social Robotics*, vol. 12(1), pp. 267–280, 2020. DOI: 10.1007/s12369-019-00560-9.
- [7] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, 2018.
- [8] W. B. Knox and P. Stone, “Interactively Shaping Agents via Human Reinforcement: The TAMER Framework,” *K-CAP*, 2009. DOI: 10.1145/1597735.1597738.

- [9] B. Argall, S. Chernova, M. Veloso, B. Browning, “A survey of robot learning from demonstration,” *Robotics and Autonomous Systems*, 57(5):469–483, 2009. DOI: 10.1016/j.robot.2008.10.024.
- [10] Aldebaran (SoftBank Robotics), *NAO6 Documentation & User Guide*. doc.aldebaran.com.

A closing thought

If the robot's reward is our approval,
it learns what pleases us — not necessarily what is good for us.

Designing for that gap — comfort versus manipulation —
may be the real research question behind adaptive robots.

Thank you — feedback welcome.

Nour Gaser & Eman Saleh

The following slides are held in reserve for discussion and deeper questions.

1. Q-learning, one concrete update (worked numerically)
2. Supervised vs. RL on NAO — a side-by-side scenario
3. Ethics deep dive — manipulation, consent, study approval
4. The reality gap — why simulation results may not transfer

Appendix 1 — One Q-learning update, step by step

Setup (Option A greeting MDP). $\alpha = 0.5$, $\gamma = 0.9$. Current estimate $Q(s_{\text{far}}, \text{closer}) = 0.0$.

1. **State** s_{far} : person in the *social* distance bin.
2. **Action** (via ϵ -greedy, exploiting): $a = \text{move closer}$.
3. **Outcome**: person stays and gives a head touch $\Rightarrow$ reward $r = +1$.
4. **Next state** $s' = s_{\text{personal}}$, where current best value is $\max_{a'} Q(s', a') = 0.4$.

Apply the update

$$Q(s_{\text{far}}, \text{closer}) \leftarrow 0.0 + 0.5 [1 + 0.9(0.4) - 0.0] = 0.5 (1.36) = 0.68$$

The estimate rises from 0.0 to 0.68: “moving closer from far *was* good here.” Repeat over episodes and the table converges to a policy that approaches to each person’s comfortable distance. [Sutton2018]

Appendix 2 — Supervised vs. RL: the same NAO task

Task: choose how close NAO stands when greeting a new person.

	Supervised learning	Reinforcement learning
Data needed	A large labeled set: (person features → “correct” distance), gathered in advance	None up front — learns from live reward (touch, sonar, dwell)
When it learns	Offline, before deployment	Online, <i>during</i> interaction
New / unseen person	Predicts from the fixed model; cannot adjust if wrong	Tries, observes feedback, corrects within the session
Failure mode	Confidently wrong on anyone unlike the training data	Some early exploration cost before convergence

Takeaway: the “right” social distance is personal and unknown in advance — exactly the setting where RL’s online adaptation beats a pre-trained classifier.

Appendix 3 — Ethics: entertainment vs. manipulation

The risk [Pollmann2023]

- Personalization that improves UX can also *steer* behavior against a person's autonomy.
- A robot optimizing “engagement” could learn manipulative tactics if the reward is naive.
- Vulnerable users (children, elderly) raise the stakes.

Our safeguards

- Reward ties to *comfort*, not raw attention-maximization.
- Transparency: the person knows the robot is adapting.
- Informed **consent** + right to stop, for every study participant.
- Data minimization: store only what the task needs.

Question for the committee

What is the required ethics / consent procedure for running the user study with participants on campus?

Appendix 4 — The reality gap: will our sim results transfer?

We prototype in simulation *first* — so we must be honest about what that does and does not prove.

Why a gap exists

- Real sonar / tactile readings are *noisy*; mocks are clean.
- Real humans behave unpredictably; scripted “users” do not.
- Joint dynamics, latency, and timing differ on hardware.

How we mitigate

- Inject noise into the mock sensors to harden the policy.
- Keep the state/action space small & discrete (robust to noise).
- Treat sim as *convergence proof*; reserve real trials for *fine-tuning*.
- Report sim and real results *separately* — never conflate them.

Honest framing: simulation shows the *algorithm* works; only the physical NAO shows the *interaction* works.